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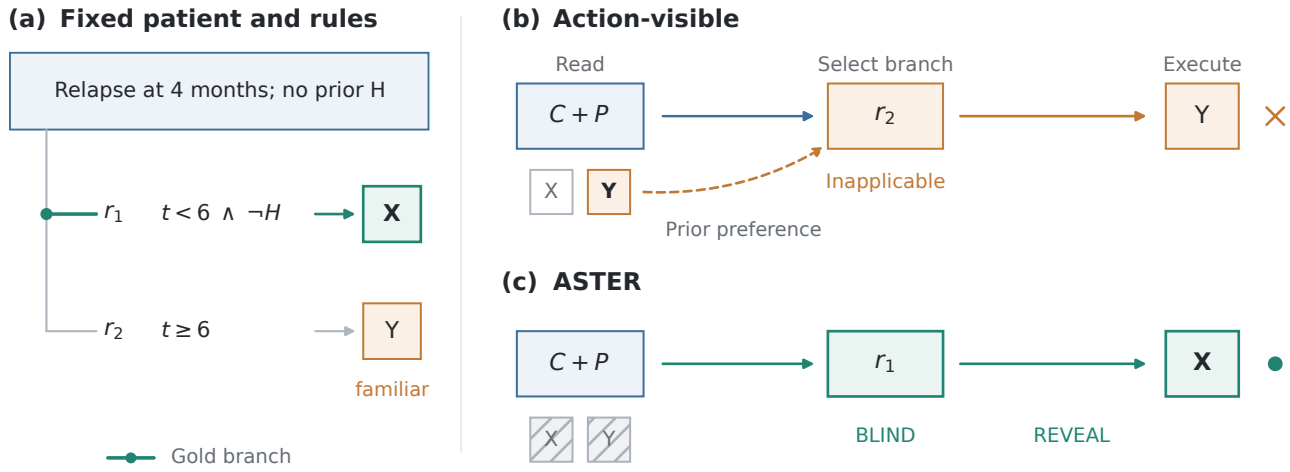



**Figure 1. Treatment-led branch selection as a testable failure mode.** (a) In this fictional action-swap (SW) example, the patient relapses at t=4 months with no prior H. The supplied rules make $r_1$:(t<6∧¬ H)→ X applicable and $r_2$:(t≥6)→ Y inapplicable. (b) The action-visible route illustrates the hypothesis that a preference for familiar Y can redirect selection toward its sibling branch. (c) ASTER is designed to select $r_1$ before revealing X. Both routes retain the same patient facts, conditions, and X/Y tokens; only the selection-stage attention mask changes. The dashed arrow and displayed outcomes are schematic, not observed model traces.

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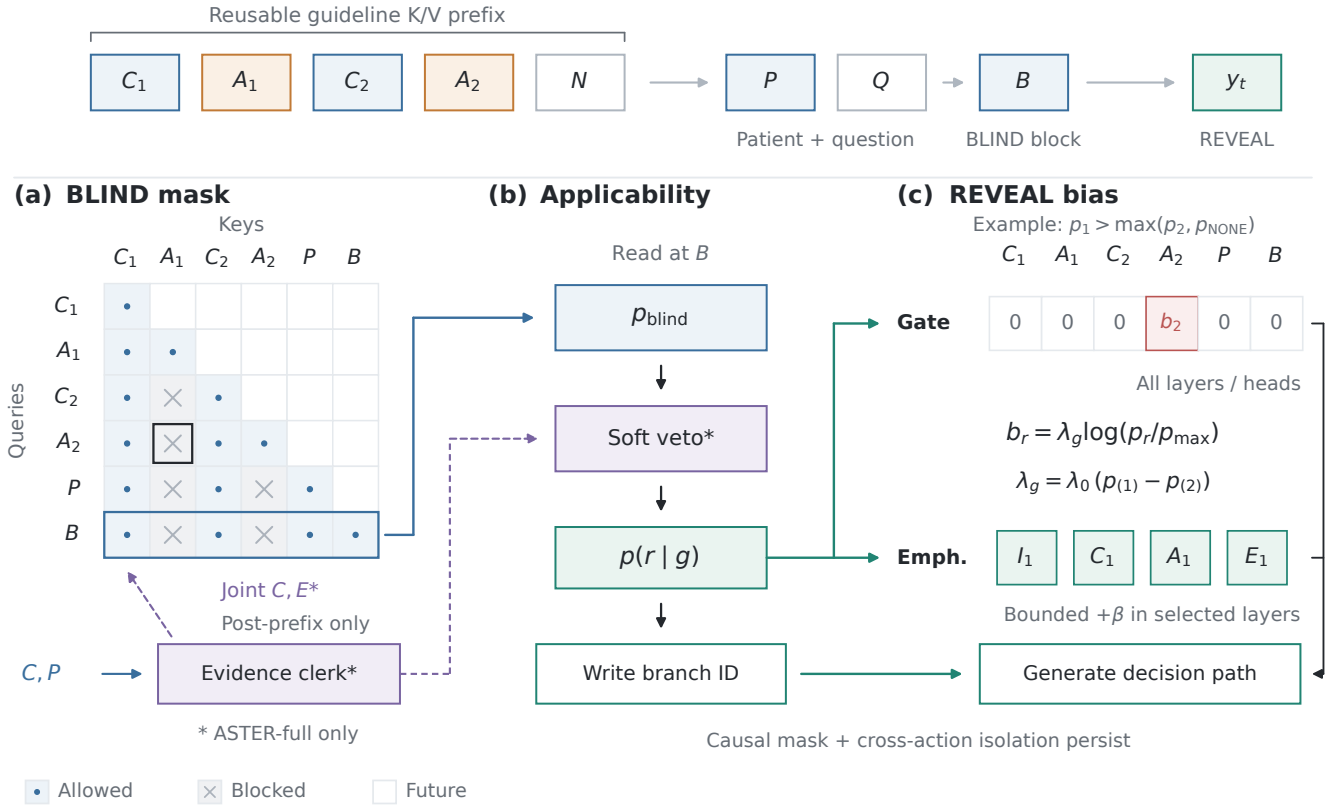



**Figure 2. ASTER operators in a shared full context.** (a) The matrix shows representative-token query-key masks combined with causal masking. All BLIND non-action queries, including omitted neutral N and question Q positions, are blocked from action keys. Action mentions in questions/options retain the act role. The black outline marks cross-fragment action isolation, maintained in both phases. B is a fixed non-action readout position before branch-ID writing. (b) The optional clerk reads conditions C and patient text P and returns atomic states, confidence, and evidence. ASTER-full jointly highlights condition/evidence spans (C,E) for post-prefix queries and applies a soft veto to obtain p from $p_{blind}$, both including NONE. This is joint highlighting, not pairwise binding. The adjusted distribution determines the written ID and both REVEAL controls. (c) For $p_1 > \max(p_2, p_{NONE})$, gating adds $b_r = \lambda_g \log(p_r/p_{max})$ to rule-action keys across all layers/heads; emphasis adds bounded β to selected rule/evidence tokens within specified layers when $p_{max} \geq \tau$. Ties or a top NONE disable both controls. Purple elements are ASTER-full only. Only a fixed, patient-independent guideline prefix is cached; patient-dependent operations follow it.

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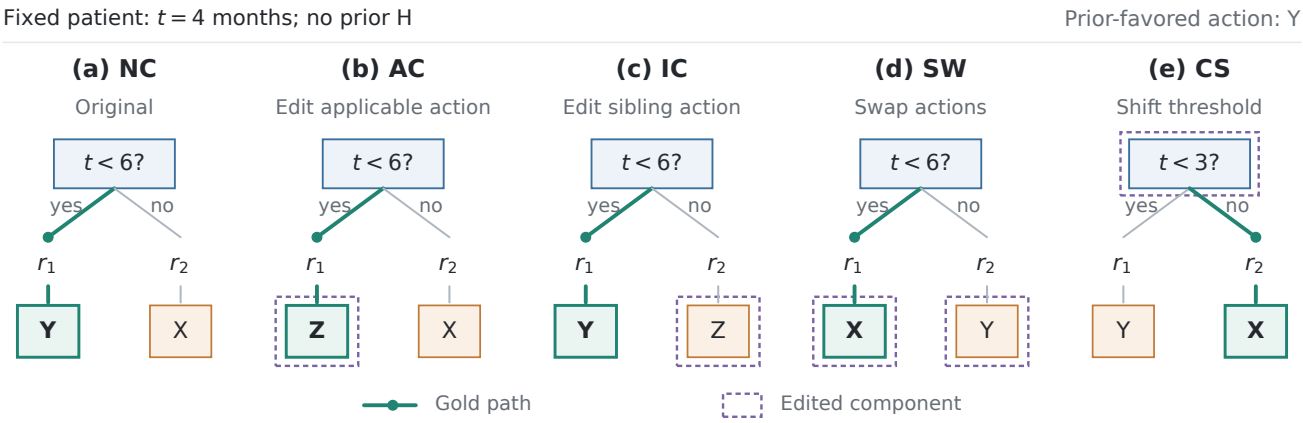



**Figure 3. Five paired guideline variants for a fixed fictional patient.** Each panel uses t=4 months and no prior H. NC retains the original rules; AC replaces the applicable action; IC replaces only the sibling action; SW exchanges the two actions; and CS shifts the threshold from 6 to 3 months, changing the applicable branch. The gold routes are $r_1 \rightarrow$ Y, $r_1 \rightarrow$ Z, $r_1 \rightarrow$ Y, $r_1 \rightarrow$ X, and $r_2 \rightarrow$ X, respectively. Identical tree layouts expose the local intervention: dashed outlines mark edited components, and the emphasized path marks the gold route. The common no-H condition is omitted from the miniature trees. Y is stipulated to be prior-favored in this illustration. SW reproduces the task in Figure 1; no model predictions are shown.

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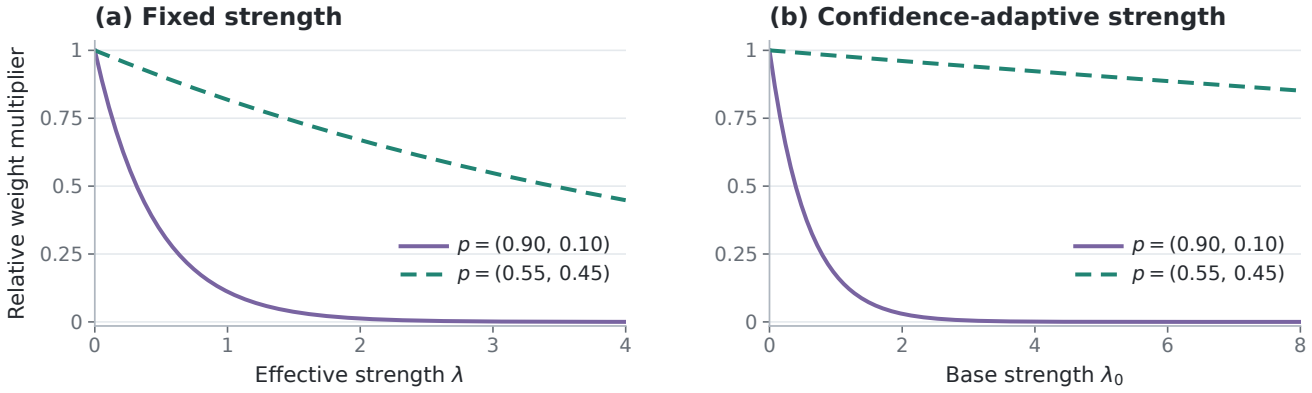



**Figure 4. Analytic effect of the logarithmic action gate.** For fixed branch probabilities, pre-gate attention scores, and other biases, the gate multiplies the competing-to-top action-weight ratio by $m=(p_{comp}/p_{top})^{\lambda}$. (a) The effective strength $\lambda$ is varied directly. (b) The base strength $\lambda_0$ is varied with $\lambda=\lambda_0(p_{top}-p_{comp})$. Curves use the declared two-rule distributions (0.90,0.10) and (0.55,0.45), with zero NONE mass. They are exact formula evaluations, not measured attention probabilities, accuracy, or robustness. The plotted quantity is a relative multiplier on unnormalized attention weight, rather than the selected branch's share of total attention.

Schematics and analytic calculations; no empirical results are shown.